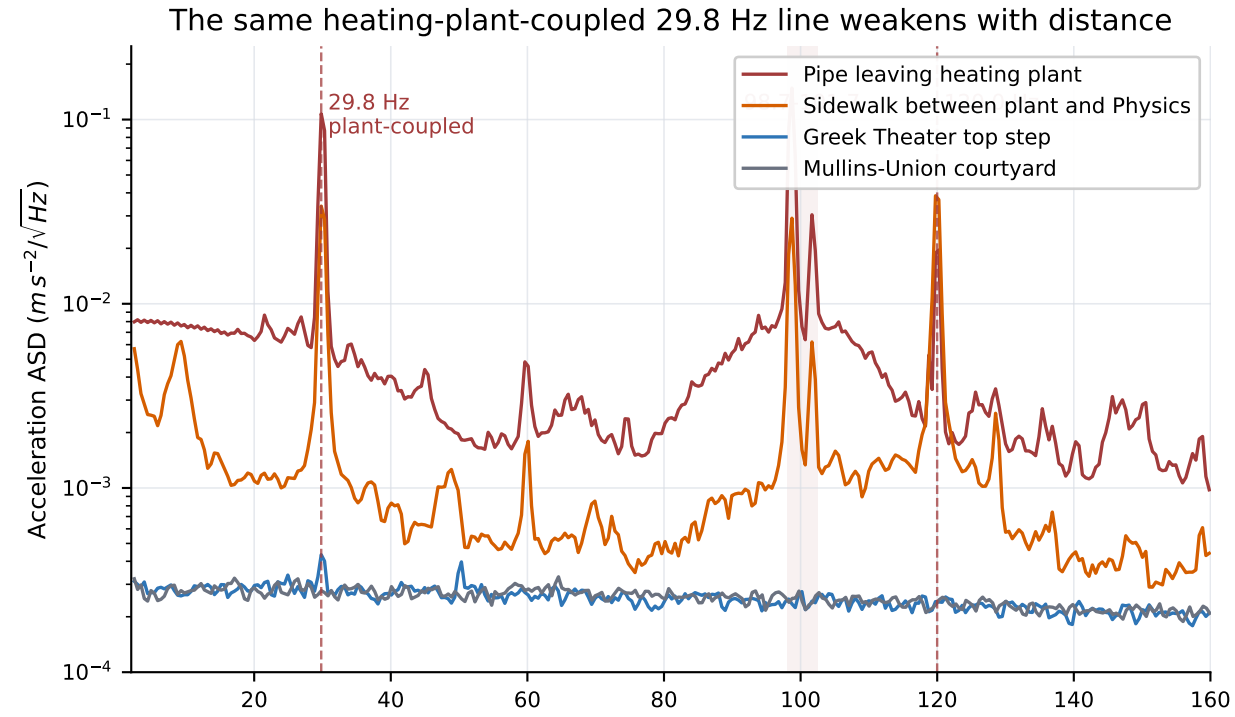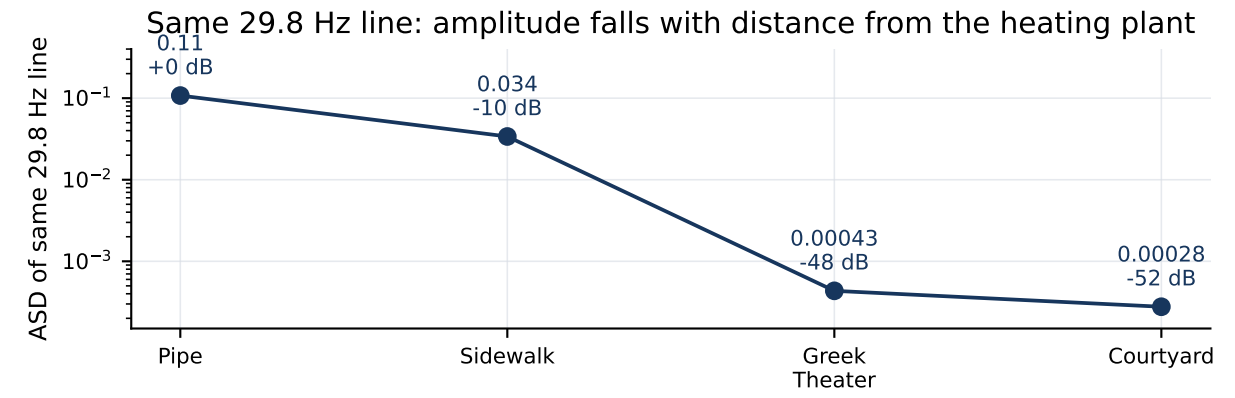Near-source and control measurements